

已知方程

$$\partial_x \partial_y \partial_z u = x^2 y^2 z^2 f'''(xyz)$$

其中

$$u = f(xyz), f(0) = 0, f'(1) = 1$$

化简为

$$f'(xyz) + 3xyz f''(xyz) + x^2 y^2 z^2 f'''(xyz) = x^2 y^2 z^2 f'''(xyz)$$

令 $xyz = t$

$$\dot{f} + 3t\ddot{f} = 0$$

观察到通解为

$$f = C_1 t^{\frac{1}{3}} + C_2$$

带入初值得到

$$u = 3(xyz)^{\frac{1}{3}}$$